Mahmoud Ahmed Mahmoud Bekhit

AI & Machine Learning Engineer | Data Scientist

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OBJECTIVE

Computer and Control Engineering graduate passionate about Machine Learning and Data Science. Experienced in developing predictive models, computer vision applications, and data-driven solutions using Python, TensorFlow, Scikit-learn, SQL, and data visualization tools. Seeking an opportunity to build scalable AI solutions and solve real-world problems through machine learning.

EDUCATION

BSc in Electrical Engineering Department of Computer and Control Engineering 2021 – 2026
Machine Learning, Data Structures, Databases, Algorithms, mathematics, logic, communication, SCADA System, Probability & Statistics, Control Systems

COURSES

GCI World – The University of Tokyo, Japan
- Data Science and Machine Learning fundamentals
- Data cleaning, preprocessing, and Exploratory Data Analysis (EDA)
- Data manipulation and visualization using Python, Pandas, NumPy, Matplotlib, and Seaborn
- Supervised and Unsupervised Machine Learning
- Feature Engineering, Model Training, and Evaluation
- Practical data analysis and predictive modeling

Data Analysis Trainee Digital Egypt Pioneers Initiative (DEPI)
- Performed data cleaning and analysis using **Microsoft Excel**
- Created reports and dashboards using **Pivot Tables**
- Wrote **SQL queries** for data extraction and analysis
- Applied **Python basics** for data analysis tasks
- Worked with real datasets to generate insights

Artificial Intelligence – Route Academy
- Python, SQL, Git
- Machine Learning & Deep Learning
- Computer Vision
- NLP
- FastAPI & Streamlit

Digital Hub with Orange
- Data analysis basics
- Data cleaning and preparation
- Exploratory Data Analysis (EDA)
- AI-driven insights generation
- Data visualization and interactive dashboards
- Prediction and automation using machine learning
- Using Excel ,Python, Power BI, Tableau

Elements of AI Minna Learn
- Understanding AI concepts and real-world applications

Cybersecurity Course Maspero (Egyptian Radio & Television Building)
- Fundamentals of system security and cyber threats

PROJECTS

AI Employment Impact Analytics & Machine Learning
Excel | SQL | Power BI | Tableau | Python | Machine Learning
- Led an end-to-end analytics project analyzing **15,000 employment records** across 8 industries, 9 countries, and 10 job roles.
- Cleaned and transformed data using Excel, SQL, and Python, and developed interactive dashboards using Power BI and Tableau.
- Designed a **7-table star-schema data warehouse** with analytical views and more than 70 business queries.

- Benchmarked 9 regression models and developed an optimized **Extra Trees Regressor**, achieving **R^2 = 0.9572**, CV R^2 = 0.9584, RMSE = 1.7517, and MAE = 1.2803.
- Generated insights related to AI adoption, automation risk, salaries, reskilling, job survival, and labor vulnerability.

Plant Disease Detection Using Deep Learning
Python | TensorFlow | Keras | CNN | Gradio | Computer Vision
- Developed a two-stage computer vision system that validates plant-leaf images before disease classification.
- Built a CNN-based classifier covering **38 healthy and diseased plant categories**, achieving **93% classification accuracy**.
- Applied data augmentation, batch normalization, dropout, early stopping, model checkpointing, and learning-rate reduction.
- Created an interactive Gradio application that displays the top three predictions with confidence scores.
- Implemented a separate leaf-detection model to reject irrelevant non-leaf images.

FC Barcelona Match Analytics Dashboard
Excel | Power Query | Power Pivot | PivotTables | Data Visualization
- Built an interactive Excel dashboard analyzing **562 FC Barcelona matches from 2015 to 2025**.
- Used Power Query for data cleaning and transformation and Power Pivot for dimensional data modeling.
- Created KPI cards, PivotTables, PivotCharts, and interactive slicers.
- Analyzed match results, goals, possession, attendance, formations, competitions, opponents, and seasonal performance.

SKILLS

Technical Skills
Programming: Python, Java, SQL
Machine Learning & Data Analysis: Scikit-learn, Regression, Classification, Model Evaluation, Feature Engineering, Data Cleaning, EDA, Data Visualization
Tools: Jupyter Notebook, Google Colab, Git & GitHub, Excel
Data Analytics: Excel, Power BI, Tableau, SQL, Power Query, Power Pivot
Libraries: Pandas, NumPy, Matplotlib, Seaborn, TensorFlow, Scikit-learn

SOFT SKILLS

- Problem Solving
- Analytical Thinking
- Communication
- Teamwork

LANGUAGES

- Arabic — Native
- English — Intermediate